

GCU Arena Energy

Assumptions

- Two basketball teams attending the game
- Each team has 12 players, 1 athletic trainer, 1 head coach and 3 assistant coaches.
- There are 36 cheerleaders per game
- The number of staff and security is 30 individuals
- Food sold at the game: fried nachos and iced soda
- There are one order of nachos and one medium drink sold per three persons
- HID lightbulb is the only electrical equipment needed to be considered for energy calculation
- The other electrical equipment contribute negligible energy releases
- The room temperature is 23°C

Data

- GCU Arena stadium has $n = 7,000$ seats
- The area of the stadium is $A = 12,500 \text{ m}^2$
- Each person releases $P = 200\text{W}$
- Total number of attendees during a basketball game: 7100 individuals
- Oil boiling temperature: 300°C
- Specific heat of a tortilla chip is $3.36 \text{ kJ}/(\text{kg} \cdot ^\circ\text{C})$
- Specific heat of water is $4.186 \text{ J}/(\text{g} \cdot ^\circ\text{C})$
- One serving size of nachos is 113 grams
- One medium cup holds 0.621 liters
- One game lasts 5400 seconds
- The stadium requires 750 Lux of brightness
- A HID light bulb has 15300 lumens per lamp
- Only 1/5 of the stadium needs lighting during the game
- Density of coke soda is 1045 g/L
- The temperature of soda with iced cube starts at 4.4°C
- The final temperature of fried tortilla chips is room temperature 23°C
- The cooking temperature of tortilla chips is the boiling temperature of boiling oil 300°C

Calculations

Lumens needed for GCU Area stadium is 1,875,000 lumens

$$\text{Lumens} = 750 \text{ Lux} \times \frac{1}{5} \times 12,500 \text{ m}^2 = 1,875,000 \text{ lumens}$$

Total number of HID light bulbs per GCU stadium is 123 lamps

$$N_{HID} = \frac{1875000 \text{ lumens}}{15300 \frac{\text{lumens}}{\text{lamp}}} = 123 \text{ lamps}$$

Total energy released by the HID lamps is 559143 kJ

$$\frac{75 \text{ lumens}}{\text{watt}} \times \frac{35 \text{ watt}}{1 \text{ lamp}} \times 123 \text{ lamps} = 322875 \text{ lumens}$$

$$(1875000 \text{ lumens} - 322875 \text{ lumens}) \times \frac{1 \text{ watt}}{75 \text{ lumens}} = 20695 \text{ watt} = 20695 \frac{\text{J}}{\text{s}}$$

$$Q_{\text{light}} = 5400 \text{ seconds} \times 20695 \frac{\text{J}}{\text{s}} = 111753 \text{ kJ}$$

Total amount of soda sold per game is 1536 kg

$$m_{\text{soda}} = \frac{1 \text{ order}}{3 \text{ persons}} \times \frac{0.621 \text{ L}}{1 \text{ order}} \times \frac{1045 \text{ g}}{1 \text{ L}} \times 7100 \text{ persons} = 1536 \text{ kg}$$

Total energy absorbed by the drinks is 119592 kJ

$$Q_{\text{drink}} = mcT = 1536 \text{ kg} \times \frac{4.186 \text{ J}}{\text{g}^\circ\text{C}} \times \frac{1000 \text{ g}}{1 \text{ kg}} \times (23 - 4.4)^\circ\text{C} = 119592 \text{ kJ}$$

Total mass of the nachos sold per game is 267 kg

$$m_{\text{nachos}} = \frac{1 \text{ order}}{3 \text{ persons}} \times \frac{0.113 \text{ kg}}{1 \text{ order}} \times 7100 \text{ persons} = 267 \text{ kg}$$

The energy released by the food is 248502 kJ

$$Q_{\text{chip}} = mcT = (267 \text{ kg}) \left(3.36 \frac{\text{kJ}}{\text{kg}^\circ\text{C}} \right) (23 - 300)^\circ\text{C} = -248502 \text{ kJ}$$

Total energy released by 7100 people in the stadium is 7668000 kJ

$$Q_{\text{people}} = 7100 \text{ people} \times \frac{200 \text{ J/s}}{\text{person}} \times 5400 \text{ s} = 7668000 \text{ kJ}$$

The main goal is to keep the temperature constant (at 23°C); therefore, the change in energy is zero. This means the air-conditioner needs to absorb the total released energy.

$$E_{\text{total}} = Q_{\text{light}} - Q_{\text{drink}} + Q_{\text{chip}} + Q_{\text{people}} = 111753 \text{ kJ} - 119592 \text{ kJ} + 248502 \text{ kJ} + 7668000 \text{ kJ}$$

$$E_{\text{total}} = 7908663 \text{ kJ}$$

$$P_{\text{total}} = \frac{7908663000 \text{ J}}{5400 \text{ s}} \times \frac{1 \text{ hp}}{746 \text{ J/s}} = 1963 \text{ horsepower}$$

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